Synchronous vs Asynchronous Execution

Project Overview

This project compares synchronous and asynchronous execution using simulated processes and communication delays. It measures the time taken by each execution model and displays the completion responses.

Programming Language: Python

Source Code

```
import time
import asyncio

# Synchronous Execution
def synchronous_task(task_name, delay):
    print(f"{task_name} started")
    time.sleep(delay)
    print(f"{task_name} completed")

def synchronous_execution():
    print("\n--- Synchronous Execution ---")
    start = time.time()

    synchronous_task("Process 1", 2)
    synchronous_task("Process 2", 3)
    synchronous_task("Process 3", 1)

    end = time.time()
    print("Synchronous Total Time:", round(end - start, 2), "seconds")


# Asynchronous Execution
async def asynchronous_task(task_name, delay):
    print(f"{task_name} started")
    await asyncio.sleep(delay)
    print(f"{task_name} completed")


async def asynchronous_execution():
    print("\n--- Asynchronous Execution ---")
    start = time.time()

    await asyncio.gather(
        asynchronous_task("Process 1", 2),
        asynchronous_task("Process 2", 3),
        asynchronous_task("Process 3", 1)
    )

    end = time.time()
    print("Asynchronous Total Time:", round(end - start, 2), "seconds")


# Main Program
if __name__ == "__main__":
    synchronous_execution()
    asyncio.run(asynchronous_execution())

    print("\n--- Execution Comparison ---")
    print("Synchronous: Tasks execute one after another.")
    print("Asynchronous: Tasks execute concurrently.")
    print("Asynchronous execution reduces waiting time.")
```

Expected Output (approximate)

```
--- Synchronous Execution ---
Process 1 started
Process 1 completed
Process 2 started
Process 2 completed
Process 3 started
```

```
Process 3 completed
Synchronous Total Time: 6.00 seconds

--- Asynchronous Execution ---
Process 1 started
Process 2 started
Process 3 started
Process 3 completed
Process 1 completed
Process 2 completed
Asynchronous Total Time: 3.00 seconds

--- Execution Comparison ---
Synchronous: Tasks execute one after another.
Asynchronous: Tasks execute concurrently.
Asynchronous execution reduces waiting time.
```

Result

The synchronous and asynchronous execution models were successfully implemented and compared. Synchronous tasks run one after another, whereas asynchronous tasks can make progress concurrently while waiting for delays. Therefore, the asynchronous version takes less time in this simulation. Actual timings may vary by computer.